

Week 7: Systems Applications & Inequalities

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Week 7 Quiz

7.1: Systems word problems (mixture, motion, break-even); choosing method

7.2: Systems of inequalities; graph feasible region; interpret solutions

7.3: Linear modeling & fit-by-eye; residual reasoning (light)

Quick Reference: Key Formulas

Mixture Problems

$$(\text{concentration}_1)(\text{volume}_1) + (\text{concentration}_2)(\text{volume}_2) = (\text{concentration}_{\text{final}})(\text{volume}_{\text{final}})$$

Motion Problems

$$\text{distance} = \text{rate} \times \text{time} \quad (d = rt)$$

Break-Even Problems

$$\text{Total Cost} = \text{Total Revenue}$$

Residuals

$$\text{Residual} = \text{Actual value} - \text{Predicted value}$$

Week 7 Quiz: Systems Applications & Inequalities

7.1 Problem 1: Mixture Problem

A chemist needs to prepare 100 mL of a 16 mg/mL saline solution.

She has solution "A", which is a 10 mg/mL solution.

She has solution "B", which is a 20 mg/mL solution.

How much of each should she mix?

7.1 Problem 2: Motion Problem

Two cars leave the same parking lot at the same time.

They travel in opposite directions.

Car A travels at 50 mph.

Car B travels at 70 mph.

After how many hours will they be 240 miles apart?

7.1 Problem 3: Break-Even Problem

A sandwich shop makes sandwiches.

The costs include a fixed cost of \$6000 per month,

plus a cost of \$2 per sandwich to produce.

The revenue from each sandwich is \$8.

How many sandwiches must be sold to break even?

7.2 Problem 4: Basic System

Graph the system of inequalities:

$$y \geq 2x - 3$$

$$y < -3x + 7$$

7.2 Problem 5: Bounded Inequalities

Graph the system of compound inequalities:

$$2 < x < 7$$

$$-1 \leq y \leq 3$$

7.2 Problem 6: Inequality Application

A bakery makes cookies and brownies.

Each cookie batch requires 2 hours of labor.

Each brownie batch requires 3 hours of labor.

The bakery has at most 30 hours of labor per day.

They must make at least 2 cookie batches per day.

Write and graph a system of inequalities.

7.3 Problem 7: Study Time and Quiz Score

A teacher tracks student study time and quiz scores.

Part A

Plot the points.

Part B

Find the equation of the line of best fit.

Part C

Calculate the residual for $x = 3$ hours of study time:

Part D

Use your model to predict the quiz score for someone who studies 8 hours.

Is this prediction reasonable? Why or why not?